AdifySkills

n8n Automation Mastery
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CLASS NOTES

🤖 Module 5 · AI Automation & Agents

DAY 17 of 25

Document AI, Vision & Voice Automation

"Invoices padho, images samjho, voice notes transcribe karo."

⚡ AAJ HUM YEH BANAYENGE

Invoice Inbox → AI Extraction → Accounting Sheet

Email mein PDF invoice aate hi n8n attachment nikaalta hai, text padhta hai, AI se vendor, invoice no, date, amount, GST nikalwata hai aur accounting sheet mein row daal deta hai.

Information Extractor | Whisper | Vision | MCP

Naam: _____ Class date: _____

2 ghante ki live class · 7:00 PM – 9:00 PM IST

🧭 Aaj ka roadmap

Ek nazar mein poori class — kya, kyun aur kitne time mein.

💡 Yeh lesson kyun zaroori hai?

Business ka asli data PDFs, images aur voice notes mein hota hai. Multimodal AI se woh sab automation ke liye usable ban jaata hai.

🎯 Aaj ke baad aap yeh kar paoge

01 PDF & invoice data extraction with AI
🌍 Bill padh ke diary mein likhna.
⚙️ Gmail attachment → Extract from File / AI → {vendor, amount, gst, date}.

02 Image analysis with vision models
🌍 Photo dekh ke batana kya hai.
⚙️ OpenAI Analyze Image: "Is product ka naam, rang, defect?"

03 Voice note transcription (Whisper)
🌍 Dadi ka voice message likh dena.
⚙️ Telegram voice → OpenAI Transcribe (Whisper) → text.

04 AI image generation in workflows
🌍 Drawing book mein poster.
⚙️ Image generate → Drive upload → social post.

05 Multi-agent workflows & orchestration
🌍 Ghar mein sabka kaam alag — ek chef, ek waiter.
⚙️ Manager agent → researcher agent + writer agent (sub-workflows).

06 MCP (Model Context Protocol) with n8n
🌍 Universal charger.
⚙️ MCP Client/Server Trigger: tools ek standard tareeke se share.

⏱ Class ka plan (2 ghante)

10m	30m	50m	20m	10m

`0:00-0:10` **Recap**
🪄 Cricket match se pehle warm-up karte ho na? Recap dimaag ka warm-up hai.

`0:10-0:40` **Concept: Multimodal AI**
🪄 Pehle robot sirf padh sakta tha. Ab woh dekh sakta hai, sun sakta hai aur drawing bhi bana sakta hai.

`0:40-1:30` **Live Build**
WhatsApp voice note → transcribe → AI answer → voice/text reply
🪄 Aaj ki kahani yaad hai — 'Robot ki aankhein, kaan aur haath'? Ab wahi kahani n8n mein banayenge. Har scene ek node hai: ek-ek block jodo aur har block ke baad dekho sab chal raha hai ya nahi.

`1:30-1:50` **Quiz + Challenge**
Module 5 quiz
🪄 Practice match — jitna yahan girenge, utna asli match mein kam girenge.

`1:50-2:00` **Q&A + Homework**
🪄 Ghar jaane se pehle bag check — sab samajh aaya? Homework diary mein likha?

🧠 Concepts — aasaan bhasha mein

Har concept ke saath ek real-life analogy. Analogy yaad rahegi to concept bhi yaad rahega.

CONCEPT 01

Information extraction

Unstructured text → fixed fields (schema).

🪄 **Analogy:** Form bharne wala clerk

CONCEPT 02

Vision

Image ko samajhna — bill, card, product photo.

🪄 **Analogy:** AI ki aankhen

CONCEPT 03

Transcription

Audio → text (Whisper).

🪄 **Analogy:** Stenographer

CONCEPT 04

MCP

Standard tareeka jisse AI agents external tools use karte hain; n8n MCP server/client dono ban sakta hai.

🪄 **Analogy:** Universal charger

🔑 Aaj ka main idea

Multimodal AI sirf text nahi — image, PDF, audio bhi samajhta hai. n8n mein file (binary) ko AI node mein bhejo aur structured data wapas lo.

- Invoice se vendor, amount, GST nikaalna
- WhatsApp voice note → text → task
- Product photo se description likhna

📖 Ek kahani se samjho: Robot ki aankhein, kaan aur haath

1 👀 **Robot ne bill dekha** = Vision / PDF extract

2 👂 **Voice note suna** = Whisper transcription

3 🎨 **Poster banaya** = Image generation

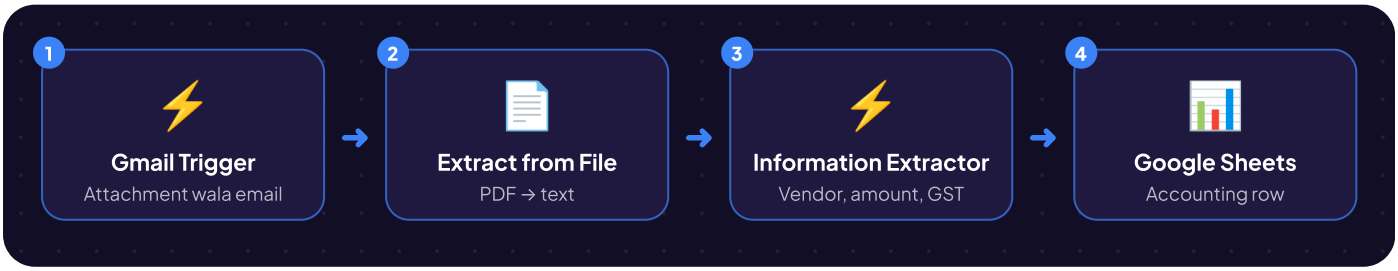
4 🤝 **Team mein kaam baanta** = Multi-agent

5 🔌 **Har tool ka ek plug** = MCP

🪄 Ek line mein: “Pehle robot sirf padh sakta tha. Ab woh dekh sakta hai, sun sakta hai aur drawing bhi bana sakta hai.”

🗺️ Aaj ka workflow: Invoice Inbox → AI Extraction → Accounting Sheet

Data left se right chalta hai. Har dabba ek n8n node hai.



💰 Isse kitna kama sakte ho?

CA firms, accounts teams (invoice extraction) — aam taur par **₹20,000 – ₹50,000**. *(Andaaza hai; client aur scope ke hisaab se badalta hai.)*

🛠 Step-by-step build — n8n kholo aur saath chalo

1. ⚡ **Gmail Trigger**
   Subject mein "invoice", has attachment
2. 📄 **Extract from File**
   PDF → text
3. ⚡ **Information Extractor**
   Schema: vendor, invoice_no, date, subtotal, gst, total
4. 🔀 **IF**
   total = subtotal + gst? (validation)
5. 📊 **Google Sheets**
   Accounting sheet append
6. ✈️ **Telegram**
   Mismatch par accountant ko alert

✅ Har step ke baad **Execute step** dabao aur output check karo. Sab green ho to workflow **Active** karo.

🔑 Setup (import ke baad)

- ☐ Connect: **Gmail (Google OAuth)**
- ☐ Connect: **OpenAI API key**
- ☐ Connect: **Google Sheets (Google OAuth)**
- ☐ "Invoices" tab headers: vendor, invoice_no, invoice_date, gstin, taxable_amount, gst_amount, total, email_from, received, status
- ☐ Agar output "output" ke andar aaye to Add Source mein {{ $json.output.vendor }} jaise map karo

✅ Test kaise karein

1. Apne aap ko ek sample invoice PDF email karo (subject mein "invoice")
2. Extractor ka output check karo — har field sahi aaya?

⚖️ Pehle vs Ab — farak dekho

Yahi farak client ko dikhana hai. Time bachta hai = paisa bachta hai.

😫 Pehle (manual)

Invoice PDF khol ke details type karna — **6 min**

6 min

⚡ Ab (n8n)

AI extraction → Sheet — **0.5 min**

0.5 min

⏱ 200 invoices/mahina = ~18 ghante bache.

⌨️ Aaj ka key snippet

Copy karke n8n mein paste karo 📋 Select karke copy karo

```
{ "vendor": "string", "invoice_no": "string", "date": "YYYY-MM-DD", "subtotal": "number", "gst": "number", "total": "number" }
```

📦 Sample data — input se output tak

📥 Input 📋 Select karke copy karo

```
{
  "text": "ABC Traders\nInvoice No: INV-2291\nDate: 02/12/2026\nSubtotal 20,000\nGST 18% 3,600\nTotal 23,600"
}
```

📤 Output 📋 Select karke copy karo

```
{
  "vendor": "ABC Traders",
  "invoice_no": "INV-2291",
  "date": "2026-12-02",
  "subtotal": 20000,
  "gst": 3600,
  "total": 23600
}
```

💬 Copy-paste prompts

ChatGPT / Claude mein seedha paste karo. PDF mein text select karke copy ho jata hai.

💬 Extractor system prompt 📋 Select karke copy karo

```
Tum ek accountant ho. Invoice text se sirf maange gaye fields nikaalo. Jo field na mile use null rakho. Amounts sirf number mein (₹ ya comma nahi). Date YYYY-MM-DD.
```

💬 Build it with ChatGPT / Claude 📋 Select karke copy karo

```
Act as an n8n expert and teacher. Help me build this workflow in n8n step by step (n8n 1.x UI):

Goal: Email mein PDF invoice aate hi n8n attachment nikaalta hai, text padhta hai, AI se vendor, invoice no, date, amount, GST nikalwata hai aur accounting sheet mein row daal deta hai.

Flow: Gmail Trigger (Attachment wala email) → Extract from File (PDF → text) → Information Extractor (Vendor, amount, GST) → Google Sheets (Accounting row)

For each node tell me: which node to add, exact settings, and the expressions to use. Then tell me how to test it and the 3 most common errors beginners hit.
```

💬 Debug helper prompt 📋 Select karke copy karo

```
I'm building the n8n workflow "Invoice Inbox → AI Extraction → Accounting Sheet" (nodes: Invoice Email, PDF to Text, Extract Invoice Data, OpenAI Model, Add Source, Accounting Sheet).
Node "<NODE NAME>" gives this error:
<PASTE ERROR>

Input JSON of that node:
<PASTE INPUT>

Explain the cause in simple words and give me the exact fix (settings or expression).
```

📥 Workflow ko n8n mein kaise laayein

1. **Workflow JSON** wale button par click karo (ya lesson page par 📋 **Copy workflow JSON** dabao).
2. Poora text copy karo (`Ctrl` + `A` , `Ctrl` + `C`).
3. n8n mein naya workflow kholo → canvas par click → `Ctrl` + `V` . Ya … menu → *Import from File*.
4. Upar wala Setup checklist poora karo aur **Test** dabao.

⚡ **Workflow JSON**
6 nodes · ready to import
`adifyskills.com/workflows/day-17-invoice-ai-extractor.json`

🌐 **Lesson page**
Copy buttons, quiz aur video links
`adifyskills.com/day/17`

🧯 Atak gaye? Yeh check karo

Error aana seekhne ka hissa hai — sab atakte hain. Neeche wale points se 90% problems solve ho jaati hain.

Agar yeh ho raha hai…	To yeh karo
Node par laal ⚠️ triangle?	Account connect nahi hua. Node kholo → Credential → "Gmail (Google OAuth)" connect karo.
Node par laal ⚠️ triangle?	Account connect nahi hua. Node kholo → Credential → "OpenAI API key" connect karo.
Node par laal ⚠️ triangle?	Account connect nahi hua. Node kholo → Credential → "Google Sheets (Google OAuth)" connect karo.
Time par apne aap nahi chala?	"Active" toggle on hai? Settings mein timezone Asia/Kolkata check karo.
AI node par 401 / 429 error?	401 = API key galat. 429 = credit khatam ya bahut jaldi requests. OpenAI billing page check karo.
"Sheet not found" / column khaali?	Sheet ID aur tab ka naam (capital/small bhi) check karo. Pehli row mein headers hone chahiye.

⚠️ Yeh galtiyan sabse common hain

- ! Scanned PDF (image) ko text extraction se padhna — OCR/vision chahiye
- ! Numbers ko string mein save (₹ aur comma)
- ! Validation skip — AI galti bhi kar sakta hai

⭐ Pro tips — time bachane wali tricks

- ★ Schema mein har field ka description likho — accuracy badhti hai
- ★ Confidence low ho to human review queue
- ★ Voice notes ke liye pehle transcribe, phir agent

Phir bhi atke? Koi baat nahi 🙂
Screenshot + error message ke saath WhatsApp par poochho →

Practice + Challenge

🏋️ Padhne se zyada banane se seekhte hain. Ek-ek level upar jaao.

🛠 Live lab

Invoice inbox: email PDF → AI extraction (vendor, amount, GST, date) → accounting sheet.

📝 Homework

Business card photo → contact details extract → CRM entry workflow.

- [ ] Homework ho gaya

BEGINNER

Business card photo → contact JSON

- [ ]

INTERMEDIATE

Voice note → todo list in Notion

- [ ]

ADVANCED

n8n workflow ko MCP server bana ke Claude/agent se call

- [ ]

🧩 Aaj ka challenge — pehle khud try karo

🎯 Agar extracted total khaali (null) aaye to sheet ki jagah aapko "manual check" email jaaye.

💡 **Hint 1**
Extraction ke baad ek check lagao.

💡 **Hint 2**
IF {{ $json.output.total }} is empty → Gmail, warna Sheet.

✅ Solution aakhri page par hai — pehle 5 minute khud socho 😉

🎯 2 minute ka self-check

Sahi option par gol ghera lagao. Answers aakhri page par — score sirf aapke liye hai.

Q1. Text se fixed fields nikaalne ka node?

(A) Summarize
(B) Sort
(C) Merge
(D) Information Extractor

Q2. Audio ko text?

(A) Transcription (Whisper)
(B) Embeddings
(C) Splitter
(D) Crypto

Q3. Scanned image PDF ke liye?

(A) XML
(B) Vision/OCR
(C) Simple text extract
(D) Sort

Q4. AI output par validation kyun?

(A) Speed
(B) Zaroori nahi
(C) AI galti kar sakta hai
(D) Fun

📖 Aaj ke naye shabd

Trigger
Jo workflow ko START karta hai — jaise alarm, naya email, ya form submit.

Agent
AI jo khud decide karta hai kaunsa tool use karna hai — sirf baat nahi, kaam bhi karta hai.

MCP

Model Context Protocol — AI tools ko ek standard plug se jodne ka tareeka.

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✍️ Meri notes

Class mein jo naya laga, jo doubt aaya — yahan likho. Kal revise karna aasaan hoga.

⏭ Kal kya seekhenge — Day 18

Error Handling, Sub-Workflows & Reliability

"Production mein workflow ka fail hona = client ka gussa."

☐ **Day 17 complete! 🎉**

Tick karo aur website par bhi ✅ dabao.

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Answer key + useful links

Pehle khud try kiya? Ab check karo. Galat hua to explanation padho — wahi asli seekh hai.

Q1 → D
Information Extractor

Q2 → A
Transcription (Whisper)

Q3 → B
Vision/OCR

Q4 → C
AI galti kar sakta hai

✅ Challenge ka solution

Extract Invoice Data → IF (total is empty) → true: Gmail "Invoice manual check karo" / false: Add Source → Sheet.

📚 Official docs

- Information Extractor ↗
- OpenAI node ↗
- MCP nodes ↗
- Extract from File ↗

▶ Videos

- n8n invoice data extraction ai ↗
- n8n whatsapp voice note ai ↗
- n8n mcp server ↗

Doubt poochho
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